Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \ldots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network. The generalized commuting cost from residence i to workplace j is

$$\tau_{ij} = \tau_{ij}(I_{ij}, n_{ij}),$$

where I_{ij} is transportation investment on link (i, j) and n_{ij} is the equilibrium commuter flow on that link. Investment reduces commuting costs and congestion raises them:

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0, \qquad \frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the commuting cost as

$$\tau_{ij} = \delta_{ij}^{\tau} \frac{n_{ij}^{\rho}}{I_{ij}^{\gamma}},$$

where $\delta_{ij}^{\tau} > 0$ captures geographic features of link (i, j) and $\rho, \gamma > 0$.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i and a workplace j to maximize utility. Preferences over the numaire good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha}\right)^{\alpha} \left(\frac{h_{ij}^R}{1-\alpha}\right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location i, $\alpha \in (0,1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each (i,j) pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j,$$

where q_i^R is the residential rent at location i, w_j is the wage at workplace j, and $\phi(N_i, L_i)$ is a monetary disutility from residing at i. The disutility captures two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \qquad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined below). I parametrize the disutility as

$$\phi(N_i, L_i) = \eta_N N_i^{1+\rho} + \eta_L L_i^{1+\rho}, \tag{1}$$

where $\eta_N, \eta_L > 0$ are congestion weights on residential crowding and worker influx respectively, and $\rho > 0$ governs the convexity of congestion costs. The marginal congestion costs are $\partial\phi/\partial N_i = (1+\rho)\eta_N N_i^{\rho}$ and $\partial\phi/\partial L_i = (1+\rho)\eta_L L_i^{\rho}$, both increasing in their arguments. Convexity ($\rho > 0$) ensures the municipal first-order condition has an interior solution: as the municipality loosens residential zoning, each additional resident raises congestion by more than the last, so the congestion cost eventually dominates the rent benefit. The ratio η_N/η_L governs the relative weight incumbents place on residential crowding versus worker influx, and is disciplined by the observed mix of residential and commercial zoning restrictions across locations. The overall scale (η_N, η_L) is calibrated to match the housing supply elasticity estimates of Baum-Snow and Han (2024), which proxy for how tightly the municipality's zoning constraint binds at the equilibrium allocation.

Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1-\alpha)Y_{ij}/q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i\, w_j}{\tau_{ij}\, (q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence i and workplace j is

$$\pi_{ij} = \frac{T_i E_j\, \tilde{V}_{ij}^{\epsilon}}{\Phi}, \qquad \Phi \equiv \sum_{i'} \Phi_{i'}, \qquad \Phi_i \equiv \sum_j T_i E_j\, \tilde{V}_{ij}^{\epsilon}, \tag{2}$$

and expected utility is

$$U = \Gamma\left(1 - \tfrac{1}{\epsilon}\right) \Phi^{1/\epsilon}. \tag{3}$$

Let N denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location i pins down the equilibrium rent:

$$q_i^R = \frac{(1-\alpha)\sum_j N\pi_{ij}\, Y_{ij}}{\bar{H}_i^R}. \tag{4}$$

1.3 Firms

At each location j, competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F:

$$Y_j = A_j L_j^{\gamma} (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0,1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j}\right)^{1-\gamma}, \qquad q_j^F = (1-\gamma) A_j \left(\frac{L_j}{H_j^F}\right)^{\gamma}.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at j gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses L_i to denote employment at location i acting as a workplace, which is this same quantity evaluated at $j = i$.

1.4 Municipal Zoning

Each municipality $g \in G$ chooses the land-use allocation $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government acts as a representative of its residents, who face a tradeoff between lower housing costs — which favor denser zoning — and lower congestion and fewer workers near their homes — which favor restrictive zoning. Formally, municipality g solves

$$\max_{\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}} \quad W^g = \sum_{i \in S^g} N_i \left(\bar{U} - \phi(N_i, L_i)\right)$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma\left(1 - \tfrac{1}{\epsilon}\right) \Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location i, subject to $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$ for each $i \in$

S^g, taking other municipalities' allocations $\{\bar{H}^R_{i'}, \bar{H}^F_{i'}\}_{i' \notin S^g}$ and transportation investment $\{I_{jk}\}$ as given.

Substituting $\bar{H}^F_i = \bar{H}_i - \bar{H}^R_i$ and differentiating $\sum_{i \in S^g} N_i \bar{U}_i$ with respect to $\bar{H}^R_i$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}^R_i}}_{\geq 0 \text{ (in-movers)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial q^R_i} \frac{\partial q^R_i}{\partial \bar{H}^R_i}}_{\geq 0 \text{ (rent benefit)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}^R_i}}_{\leq 0 \text{ (congestion cost)}} + N_i \underbrace{\left(- \frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}^F_i} \right)}_{\geq 0 \text{ (less worker influx)}} = 0. \tag{5}$$

The first term is positive: looser zoning lowers rents, attracting additional residents from the rest of the metro area; each in-mover contributes $\bar{U}_i$ to the municipality's objective. The second term is positive: lower rents raise effective income for all N_i existing residents. The third term is negative: in-movers raise N_i, worsening congestion for all current residents through ϕ. The fourth term is positive: shifting land from commercial to residential reduces employment L_i and the associated worker congestion. The municipality sets $\bar{H}^R_i$ where the benefit of attracting residents and lowering rents is offset by the congestion they impose.

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{jk}\}_{j,k \in S}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}^R_i, \bar{H}^F_i\}$ as given:

$$\max_{\{I_{jk}\}} N \cdot \Gamma\big(1 - \tfrac{1}{\epsilon}\big)\, \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{j,k} \delta_{jk}\, I_{jk} \leq K, \qquad I_{jk} \geq 0,$$

where δ_{jk} is the per-unit cost of investment on link (j,k) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. The interior first-order condition for link (j,k) is

$$\frac{N}{\epsilon} \Gamma\big(1 - \tfrac{1}{\epsilon}\big)\, \Phi^{1/\epsilon - 1} \frac{\partial \Phi}{\partial I_{jk}} = \mu\, \delta_{jk}. \tag{6}$$

The left-hand side is the marginal social benefit of investment on link (j,k): lower commuting costs raise $\tilde{V}_{ij}$ for all commuters on that link, increasing Φ and hence aggregate welfare. The authority equates the marginal benefit per dollar of investment across all active links.

References

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[4] Vincent Rollet and Laura Weiwu. *Measuring Winners and Losers from Increasing Housing Supply*. en. Research rep. 2025.